

Non-vanishing

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1 Non-vanishing

For terminal threefolds, by Miyaoka [Miy88]:

Theorem 1.1. *Let X be a normal projective threefold defined over the complex numbers. Assume that the singular points of X are all terminal and that the canonical divisor $K_X \in \text{Pic}(X) \otimes \mathbb{Q}$ is numerically effective. Then $\kappa(X) \geq 0$, i.e., there is a positive integer m such that mK_X is integral with $|mK_X| \neq \emptyset$.*

For klt threefold pairs, by [KMM94]:

Theorem 1.2 (Log abundance). *Let the pair (X, B) consist of a threefold X and boundary B , such that $K_X + B$ is nef and log canonical. Then $|m(K_X + A)|$ is basepoint free for some m .*

Let X be a minimal threefold whose canonical divisor is numerically trivial (for example, a Calabi-Yau manifold). Then any effective nef divisor is semiample.

Theorem 1.3. *Let the pair (X, B) consist of a threefold and boundary B such that $K_X + B$ is nef and Kawamata log terminal. Then $\kappa(K_X + B) \geq 0$.*

Anti-canonical non-vanishing, by [LMPTX23]:

Theorem 1.4. *Let (X, B) be a projective log canonical pair of dimension 3 such that $-(K_X + B)$ is nef. Assume that X is $\mathbb{Q}$ -factorial or that it has rational singularities. Then the following hold.*

1. *The numerical class of $-(K_X + B)$ is effective.*
2. *If X is uniruled, then the numerical class of any nef Cartier divisor on X is effective.*

Corollary 1.5. *Let X be a projective, $\mathbb{Q}$ -factorial, log canonical pair of dimension 3, let $\epsilon \in \{-1, 1\}$, and assume that $\epsilon(K_X + B)$ is nef. Then the numerical class of $\epsilon(K_X + B)$ is effective.*

For positive characteristic, by [XZ19]:

Theorem 1.6. *Let X be a terminal projective minimal threefold over an algebraically closed field k of characteristic $p > 5$. Then $\kappa(X, K_X) \geq 0$; that is, nonvanishing holds for X .*

2 Non-vanishing and abundance

If dlt extension and non-vanishing hold, then abundance holds, by [DHP13].

If $\nu(X) = 1$ and non-vanishing holds, then abundance holds, by [LX25].

3 KM non-vanishing

For all dimensions with additional conditions. By [KM98],

Theorem 3.1. *If X is a projective variety with klt singularities and L is a nef Cartier divisor on X such that $L - (K_X + \Delta)$ is nef and big for some effective $\mathbb{Q}$ -divisor Δ such that (X, Δ) is klt, then $H^0(X, L) \neq 0$.*

By [BP11]

Theorem 3.2. *Let $(X/Z, B)$ be a klt pair where B is big/ Z . If $K_X + B$ is pseudo-effective/ Z , then $(X/Z, B)$ is effective. That is, there is an $\mathbb{R}$ -divisor $M \geq 0$ such that $K_X + B \equiv M/Z$.*

4 Others

For Kahler threefolds, see [HLL25]:

References

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